

本次课内容

氢原子 (2)

1. 模型分析

2. 求解氢原子 (内部) 能级

3. 氢原子波函数

§ 3.4 本征函数系的一般性质



预习思考题：

1. 氢原子内部能量 E_n 和角动量 L 分别是什么具体的物理量？
2. 氢原子径向函数 $R(r)$ 的方程如何得到，径向运动与能量量子化是什么关系？如果 r 固定（如同刚性转子），能级有何不同？
3. 氢原子波函数有哪些特点，“电子云”代表什么？为什么电子云空间分布有方向性？

回顾与思考

1. “束缚” 导致能量量子化

(1) 无限深势阱（一维、三维）

(2) 谐振子（一维、三维）

(3) 一维有限深势阱

2. 三维问题的分离变量法

3. 粒子位置的空间概率分布

（定态能量确定，位置不确定）

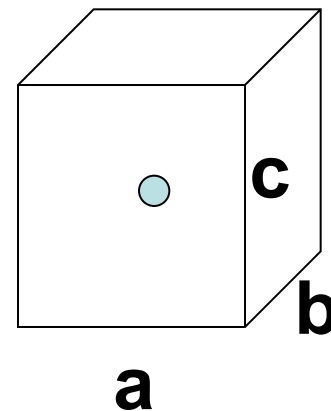


4. 氢原子为什么有原子半径？（基态下电子与核为什么没有紧贴在一起）



回顾与思考

- 求解三维无限深方势阱（考虑一个粒子被囚禁在一个长方体盒子内，盒内 $U=0$ ，盒外 $U=+\infty$ ，求能量本征值与本征函数）



对无穷大势阱，粒子不能到达盒外，即盒外波函数为0

提示:

分别求出X, Y, Z
后代回

设 $\psi(x, y, z) = X(x)Y(y)Z(z)$

(三维) 定态薛定谔方程, 对盒内区域, 有

$$-\frac{\hbar^2}{2\mu} \left(\frac{d^2}{dx^2} + \frac{d^2}{dy^2} + \frac{d^2}{dz^2} \right) \psi = E\psi(x, y, z) \quad (0 < x < a, 0 < y < b, 0 < z < c)$$

$$-\frac{\hbar^2}{2\mu} (X''YZ + XY''Z + XYZ'') = E\psi(x, y, z)$$

两边同除以XYZ得

$$-\frac{\hbar^2}{2\mu} (X''/X + Y''/Y + Z''/Z) = E$$

以上每一项必须是常数 (否则等式不能成立)

$$\text{设 } X''/X = -k_x^2; \quad Y''/Y = -k_y^2; \quad Z''/Z = -k_z^2$$

$$(X'' \equiv \frac{d^2}{dx^2} X)$$

答案

(选择原点在顶点处)

$$\psi_{n,m,l}(\mathbf{x},\mathbf{y},\mathbf{z}) = \sqrt{\frac{8}{abc}} \sin\left(\frac{n\pi}{a}x\right) \sin\left(\frac{m\pi}{b}y\right) \sin\left(\frac{l\pi}{c}z\right)$$

$$E_{n,m,l} = \frac{\hbar^2}{2\mu}(k_x^2 + k_y^2 + k_z^2) = \frac{\hbar^2}{2\mu} \left[\left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2 + \left(\frac{l\pi}{c}\right)^2 \right]$$

$(n, l, m = 1, 2, 3, \dots)$

(μ 是粒子质量)

1. 模型的建立 回顾

1.1 经典模型:

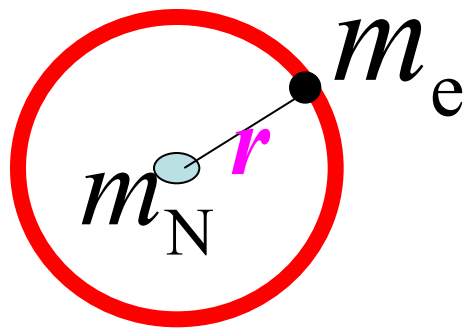
$$H = \frac{1}{2} m_N \dot{\vec{r}}_N^2 + \frac{1}{2} m_e \dot{\vec{r}}_e^2 + U(|\vec{r}_e - \vec{r}_N|) \quad (1)$$

$$\vec{r} = \vec{r}_e - \vec{r}_N \text{ (相对坐标)}$$

引入质心坐标 $\vec{R}$ 和相对坐标 $\vec{r}$

$$\begin{cases} \vec{R} = \frac{m_N \vec{r}_N + m_e \vec{r}_e}{m_N + m_e} \\ \vec{r} = \vec{r}_e - \vec{r}_N \end{cases} \Rightarrow \begin{cases} \vec{r}_N = \vec{R} - \frac{m_e}{m_e + m_N} \vec{r} \\ \vec{r}_e = \vec{R} + \frac{m_N}{m_e + m_N} \vec{r} \end{cases}$$

代入 (1)
 $\dot{\vec{R}} \cdot \dot{\vec{r}}$ 项抵消
得下页结果



回顾

$$H = \frac{1}{2} M \dot{\mathbf{R}}^2 + \frac{1}{2} \mu \dot{\mathbf{r}}^2 + U(\mathbf{r})$$

(质心运动)

(相对运动)

定义 $\vec{p}_R = M \dot{\mathbf{R}}$; $\vec{p} = \mu \dot{\mathbf{r}}$

$$H = \frac{\vec{p}_R^2}{2M} + \frac{\vec{p}^2}{2\mu} + U(r)$$

(质心运动)

(相对运动)

$$M = m_N + m_e$$

$$\mu = \frac{m_N m_e}{M} \text{ (约化质量)}$$

$$\frac{1}{\mu} = \frac{1}{m_N} + \frac{1}{m_e}$$

$\vec{p} = \mu \dot{\vec{r}}$ 的物理含义 回顾

$$\left\{ \begin{array}{l} \vec{r}_N = \vec{R} - \frac{m_e}{m_e + m_N} \vec{r} \Rightarrow \text{质心系中原子核坐标} \\ \vec{r}_N' = \vec{r}_N - \vec{R} = -\frac{m_e}{m_e + m_N} \vec{r}, \\ \vec{r}_e = \vec{R} + \frac{m_N}{m_e + m_N} \vec{r} \Rightarrow \text{质心系中电子坐标} \\ \vec{r}_e' = \vec{r}_e - \vec{R} = \frac{m_N}{m_e + m_N} \vec{r} \end{array} \right.$$

$$\text{质心系中电子动量 } \vec{p}_e' = m_e \dot{\vec{r}}_e' = \frac{m_e m_N}{m_e + m_N} \dot{\vec{r}} = \mu \dot{\vec{r}} = \vec{p}$$

$$\text{质心系原子核动量 } \vec{p}_N' = m_N \dot{\vec{r}}_N' = -\frac{m_e m_N}{m_e + m_N} \dot{\vec{r}} = -\mu \dot{\vec{r}} = -\vec{p}$$

氢原子 (2)

1. 模型分析

1.1 经典模型 (回顾上次课) :

上次课已证: $\vec{p}_e' = -\vec{p}_N' = \vec{p} (= \mu \dot{\vec{r}})$

$$\mu = \frac{m_N m_e}{m_N + m_e}$$

所以质心系内部的动能 $\frac{p_e'^2}{2m_e} + \frac{p_N'^2}{2m_N} = \frac{p^2}{2\mu}$

$$H = \frac{\vec{p}_R^2}{2M} + \frac{\vec{p}^2}{2\mu} + U(r)$$

(质心运动) (内部运动)

$$\frac{1}{\mu} = \frac{1}{m_N} + \frac{1}{m_e}$$

↓
两体运动等效为质量为 μ 的单体运动

类似地， $\vec{L} = \vec{r} \times \vec{p}$ 的物理含义是质心参考系中电子和原子核绕质心的总角动量

证明：由前面的结果 $\vec{p} = \vec{p}_e' = -\vec{p}_N'$ ，
有
$$\begin{aligned}\vec{L} &= \vec{r} \times \vec{p} = (\vec{r}_e - \vec{r}_N) \times \vec{p} = (\vec{r}_e' - \vec{r}_N') \times \vec{p} \\ &= \vec{r}_e' \times \vec{p}_e' + \vec{r}_N' \times \vec{p}_N' \\ &= \vec{L}_e' + \vec{L}_N'\end{aligned}$$

(即质心系中电子和原子核总角动量)

等效质量为 μ 的单粒子运动代表两体运动

1.2 量子模型:

量子化: $\vec{p}_R = -i\hbar\nabla_R$

量子化: $\vec{p} = -i\hbar\nabla$

$$H = H = \frac{p_R^2}{2M} + \frac{p^2}{2\mu} + U(r)$$

(质心运动) (相对运动)

$$= -\frac{\hbar^2}{2M}\nabla_R^2 - \frac{\hbar^2}{2\mu}\nabla^2 + U(r)$$

$$(\nabla_R^2 = \frac{\partial^2}{\partial X^2} + \frac{\partial^2}{\partial Y^2} + \frac{\partial^2}{\partial Z^2};$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2})$$

$$i\hbar \frac{\partial}{\partial t} \Psi(\vec{r}, \vec{R}, t) = H\Psi$$

↓

此波函数的物理含义是什么?

答案: 代表系统整体状态 (不是单粒子状态)。

求解薛定谔方程:

设分离变量特解: $\Psi(\vec{r}, \vec{R}, t) = \tilde{\Psi}(\vec{r}, \vec{R}) e^{-\frac{i}{\hbar} E_{\text{总}} t}$

定态薛定谔方程 $H \tilde{\Psi}(\vec{r}, \vec{R}) = E_{\text{总}} \tilde{\Psi}(\vec{r}, \vec{R})$

$$\left[-\frac{\hbar^2}{2M} \nabla_{\vec{R}}^2 - \frac{\hbar^2}{2\mu} \nabla^2 + U(r) \right] \tilde{\Psi}(\vec{r}, \vec{R}) = E_{\text{总}} \tilde{\Psi}(\vec{r}, \vec{R}) \quad (2)$$

继续分离变量: 设 $\tilde{\Psi}(\vec{r}, \vec{R}) = \psi(\vec{r}) \phi(\vec{R})$, 代入(2)得

代表一种特殊状态 $\downarrow$ 内部 $\downarrow$ 质心

$$\left[-\frac{\hbar^2}{2M} \nabla_{\vec{R}}^2 - \frac{\hbar^2}{2\mu} \nabla^2 + U(r) \right] \psi(\vec{r}) \phi(\vec{R}) = E_{\text{总}} \psi(\vec{r}) \phi(\vec{R})$$

两边同除以 $\psi(\vec{r}) \phi(\vec{R})$,

$$\frac{-\frac{\hbar^2}{2M} \nabla_{\vec{R}}^2 \phi(\vec{R})}{\phi(\vec{R})} + \frac{-\frac{\hbar^2}{2\mu} \nabla^2 \psi(\vec{r})}{\psi(\vec{r})} + U(r) = E_{\text{总}}$$

↪ E

$$\left[-\frac{\hbar^2}{2\mu} \nabla^2 + U(r) \right] \psi(\vec{r}) = E \psi(\vec{r}) \text{ (相对运动)}$$

(原子内部运动)

$$-\frac{\hbar^2}{2M} \nabla_{\vec{R}}^2 \phi(\vec{R}) = (E_{\text{总}} - E) \phi(\vec{R}) \text{ (质心运动) (自由)}$$

这是质心自由运动情形 (其它情形物理内容更丰富) 。

2. 求解氢原子（内部）能级

$$\left[-\frac{\hbar^2}{2\mu} \nabla^2 + U(r) \right] \psi = E \psi(\vec{r}),$$

动能



$$-\frac{\hbar^2}{2\mu} \nabla^2 = -\frac{\hbar^2}{2\mu r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{2\mu r^2} \hat{L}^2$$

径向动能

转动动能



r 变化, 不是刚性转子

解释如下

附：角动量平方和拉普拉斯算符的关系

$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right].$$
$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}$$

$$= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) - \frac{\hat{L}^2}{\hbar^2 r^2}$$

思考题：动能是量子化的吗？ 答：否

$$\rightarrow -\frac{\hbar^2}{2\mu} \nabla^2 = -\frac{\hbar^2}{2\mu r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{2\mu r^2} \hat{L}^2$$

$\frac{\tilde{p}^2}{2\mu}$ 连续谱

求解相对运动方程

$$\left[-\frac{\hbar^2}{2\mu} \nabla^2 + U(r) \right] \psi = E \psi(\vec{r}),$$

动能



$$\left[-\frac{\hbar^2}{2\mu r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + U(r) + \frac{1}{2\mu r^2} \hat{L}^2(\theta, \phi) \right] \psi = E \psi(\vec{r}).$$

径向动能

转动动能

移项,两边同乘 r^2

$$\left[\underbrace{-\frac{\hbar^2}{2\mu r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right)}_{\text{径向}} + U(r) - E \right] \psi r^2 = - \underbrace{\frac{1}{2\mu} \hat{L}^2(\theta, \phi)}_{\text{角向}} \psi$$

已知 $\hat{L}^2$ 本征函数 $Y(\theta, \phi)$ 满足 $\hat{L}^2 Y = l(l+1)\hbar^2 Y$

分离变量，设 $\psi = R(r)Y(\theta, \phi)$,

$$-\frac{\hbar^2}{2\mu r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \left[U(r) + \frac{l(l+1)\hbar^2}{2\mu r^2} \right] R = ER(r).$$

库仑势 $U(r) = -k_1 \frac{Ze^2}{r}$

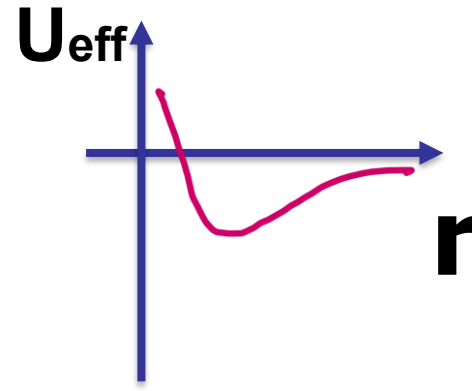
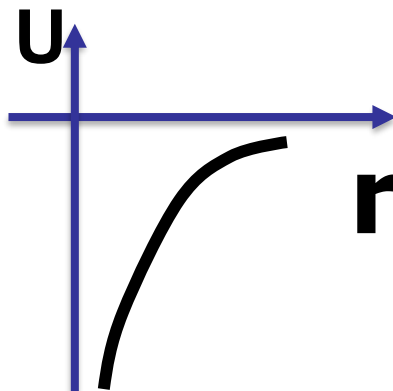
(氢原子 $Z=1$)

$$k_1 = \begin{cases} (4\pi\epsilon_0)^{-1}, & (\text{SI}) \\ 1, & (\text{CGS}) \end{cases}$$

再引入变换 $R(r) = \frac{u(r)}{r}$ ($r \rightarrow 0$ 时 $u \rightarrow 0$)

得
$$-\frac{\hbar^2}{2\mu} \frac{d^2 u}{dr^2} + \left[-k_1 \frac{Ze^2}{r} + \frac{l(l+1)\hbar^2}{2\mu r^2} \right] u = Eu(r).$$

$$U_{\text{eff}} = U(r) + \frac{l(l+1)\hbar^2}{2\mu r^2}$$



以下只分析束缚态,

$$E < U(\infty) = 0 \quad (E < U_{\text{eff}}(\infty))$$

$$-\frac{\hbar^2}{2\mu} \frac{d^2 u}{dr^2} + \left[-k_1 \frac{Ze^2}{r} + \frac{l(l+1)\hbar^2}{2\mu r^2} \right] u = \underline{Eu(r)}.$$

• 定义 $\rho = \alpha r$ $\left(\alpha = \frac{\sqrt{8\mu|E|}}{\hbar} \right),$

无量纲化

$$\beta = \frac{2\mu k_1 Ze^2}{\alpha \hbar^2} = \frac{k_1 Ze^2}{\hbar} \sqrt{\frac{\mu}{2|E|}}$$

$$\frac{d^2 u}{d\rho^2} + \left[-\frac{1}{4} + \frac{\beta}{\rho} - \frac{l(l+1)}{\rho^2} \right] u = 0. \quad (1)$$

(假设 $E < 0$ (束缚态))

两种极限情况

$\rho \rightarrow \infty$, 保留 (1) 的[...]中第1项

$$\frac{d^2 u}{d\rho^2} - \frac{1}{4} u \approx 0, \quad u(\rho) \rightarrow \exp\left(-\frac{1}{2}\rho\right);$$



E < 0

$\rho \rightarrow 0$ (保留 (1) 的[...]中第3项)

$$\frac{d^2 u}{d\rho^2} - \frac{l(l+1)}{\rho^2} u \approx 0, \quad u(\rho) \rightarrow \rho^{l+1}.$$




一般解

设 $u(\rho) = \overset{\rho \rightarrow 0}{\downarrow} \rho^{l+1} \exp\left(\overset{\rho \rightarrow \infty}{\downarrow} -\frac{1}{2}\rho\right) \underline{v(\rho)},$

代入方程 $\frac{d^2 u}{d\rho^2} + \left[-\frac{1}{4} + \frac{\beta}{\rho} - \frac{l(l+1)}{\rho^2} \right] u = 0.$

得 $\frac{d^2 v}{d\rho^2} + \left[\frac{2(l+1)}{\rho} - 1 \right] \frac{dv}{d\rho} + \frac{\beta - (l+1)}{\rho} v = 0$

求解 $\frac{d^2 v}{d\rho^2} + \left[\frac{2(l+1)}{\rho} - 1 \right] \frac{dv}{d\rho} + \frac{\beta - (l+1)}{\rho} v = 0$

设 $v = \sum_{k=0}^{\infty} c_k \rho^k$, 代入以上方程, 得

$$\sum_{k=2}^{\infty} c_k k(k-1) \rho^{k-2} + \left[\frac{2(l+1)}{\rho} - 1 \right] \sum_{k=1}^{\infty} c_k k \rho^{k-1} + \frac{\beta - (l+1)}{\rho} \sum_{k=0}^{\infty} c_k \rho^k = 0,$$

比较 ρ 的同类项系数，得

$$c_{k+1} = \frac{k - [\beta - (l + 1)]}{(k + 1)k + 2(k + 1)(l + 1)} c_k$$

截断条件：必须存在某个 n_r ，取 $k=n_r$ 时 $c_{n_r+1} = 0$
عزوة

(之后的系数当然也就为0) ，

这要求： $n_r - [\beta - (l + 1)] = 0$

造成能量量子化！

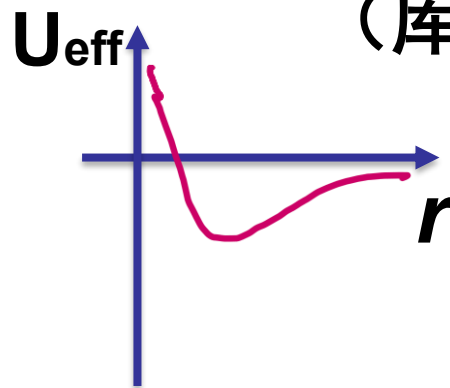
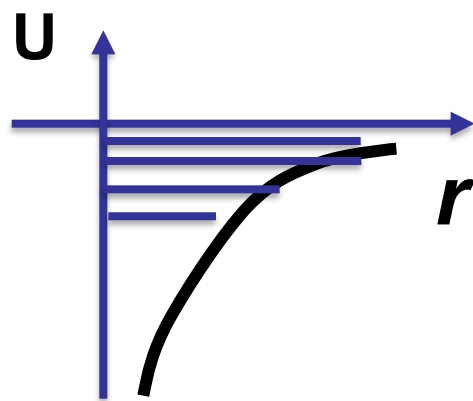
$$\beta = n_r + l + 1 \equiv n, \quad (n_r = 0, 1, 2, \dots)$$

$$\beta = \frac{k_1 Z e^2}{\hbar} \sqrt{\frac{\mu}{2|E|}} = n \quad (\text{氢原子 } Z=1)$$

$$= n_r + l + 1 (n_r = 0, 1, 2, \dots)$$

$$l = n - n_r - 1 \Rightarrow l = 0, 1, \dots, n - 1$$

得到能级 $E_n = -\frac{\mu k_1^2 Z^2 e^4}{2\hbar^2} \frac{1}{n^2} (n = 1, 2, 3, \dots)$



(库仑势造成能量量子化)

$$k_1 = \begin{cases} (4\pi\epsilon_0)^{-1}, & (\text{SI}) \\ 1, & (\text{CGS}) \end{cases}$$

$E > 0$ 非束缚态（连续谱）

E_∞

E_2

E_1

$E < 0$ 束缚态（离散谱）

束缚态能级 $E_n = \frac{E_1}{n^2}$, $E_1 = -\frac{\mu k_1^2 e^4}{2\hbar^2} \approx -13.6 \text{ eV}$

$$k_1 = \begin{cases} (4\pi\epsilon_0)^{-1}, & (\text{SI}) \\ 1, & (\text{CGS}) \end{cases}$$

（n很大的束缚态——里德堡态，有奇特性质）

3. 氢原子波函数

束缚（定）态波函数（空间部分）

$$\psi_{nlm}(r, \theta, \phi) = R_{nl}(r) Y_{lm}(\theta, \phi)$$



径向
函数



角向
(球谐)
函数

径向函数

$$R_{nl}(r) = \frac{u_{nl}(r)}{r}, \quad \rho = \alpha r \quad \left(\alpha = \frac{\sqrt{8\mu|E|}}{\hbar} \right),$$

$$u_{nl}(r) = N_{nl} \rho^{l+1} v_{nl}(\rho) \exp\left(-\frac{1}{2}\rho\right),$$

$$\int_0^\infty dr R_{nl}^2(r) r^2 = 1$$

$$v_{nl} = \sum_{k=0}^{n-l-1} c_k \rho^k$$

$$c_{k+1} = \frac{k - [\beta - (l+1)]}{(k+1)k + 2(k+1)(l+1)} c_k$$

级数递推公式归纳为： $v_{nl}(\rho) = L_{n-l-1}^{2l+1}(\rho)$

$$L_m^k(x) = \frac{e^x}{m! x^k} \frac{d^m}{dx^m} \left(\frac{x^{m+k}}{e^x} \right) \quad \text{拉盖尔多项式}$$

氢原子 波函数

$$\psi_{nlm}(r, \theta, \phi) = \sqrt{\left(\frac{2}{na}\right)^3 \frac{(n-l-1)!}{2n(n+l)!}} \times \left(\frac{2r}{na}\right)^l e^{-\frac{r}{na}} L_{n-l-1}^{2l+1}\left(\frac{2r}{na}\right) Y_{lm}(\theta, \phi)$$

$$L_m^k(x) = \frac{e^x}{m! x^k} \frac{d^m}{dx^m} \left(\frac{x^{m+k}}{e^x} \right) \quad a = \frac{\hbar^2}{\mu k_1 e^2} \approx 0.53 \text{ \AA} \quad \square \quad \text{Bohr半径}$$

$$Y_{lm}(\theta, \phi) = (-1)^m \sqrt{\frac{(2l+1)}{4\pi} \frac{(l-|m|)!}{(l+|m|)!}} P_l^m(\cos \theta) \exp(im\phi),$$

$$P_l^m(w) = \frac{1}{2^l l!} (1-w^2)^{|m|/2} \frac{d^{l+|m|}}{dw^{l+|m|}} (w^2-1)^l$$

举例 $\psi_{nlm}(r, \theta, \phi) = R_{nl}(r)Y_{lm}(\theta, \phi)$

$$\psi_{100} = \sqrt{\frac{1}{\pi a^3}} \exp\left(-\frac{r}{a}\right) = \sqrt{\frac{4}{a^3}} \exp\left(-\frac{r}{a}\right) Y_{00}(\theta, \phi),$$

$$\psi_{200} = \sqrt{\frac{1}{2a^3}} \left(1 - \frac{r}{2a}\right) \exp\left(-\frac{r}{2a}\right) Y_{00}(\theta, \phi),$$

$$Y_{00} = \sqrt{\frac{1}{4\pi}},$$

$$\psi_{21m} = \sqrt{\frac{1}{6a^3}} \left(\frac{r}{2a}\right) \exp\left(-\frac{r}{2a}\right) Y_{1m}(\theta, \phi). \quad (m = 1, 0, -1)$$

$$a = \frac{\hbar^2}{\mu k_1 e^2} \approx 0.53 \text{ \AA} \quad \square \quad \text{Bohr半径} \quad \text{位置不确定度}$$

$$k_1 = \begin{cases} (4\pi\epsilon_0)^{-1}, & (\text{SI}) \\ 1, & (\text{CGS}) \end{cases}$$

$$\left\{ \begin{array}{l}
 Y_{00} = \sqrt{\frac{1}{4\pi}}, \quad (\text{偶宇称}) \\
 Y_{10} = \sqrt{\frac{3}{4\pi}} \cos \theta = \sqrt{\frac{3}{4\pi}} \frac{z}{r} \quad (\text{奇宇称}) \\
 Y_{1,\pm 1} = \mp \sqrt{\frac{3}{8\pi}} \sin \theta \exp(\pm i\phi) = \mp \sqrt{\frac{3}{8\pi}} \sin \theta (\cos \phi \pm i \sin \phi) \\
 = \sqrt{\frac{3}{8\pi}} \frac{(\mp x - iy)}{r} \quad (\text{奇宇称})
 \end{array} \right.$$

$$Y_{lm}(-x, -y, -z) = (-1)^l Y_{lm}(x, y, z)$$

定态波函数的宇称性质

$$\psi_{nlm}(r, \theta, \varphi) = R_{nl}(r) Y_{lm}(\theta, \varphi),$$

$$\vec{r} \rightarrow -\vec{r}$$

偶

$$(-1)^l$$

$$r = \sqrt{x^2 + y^2 + z^2} \quad (\text{偶函数})$$

$$R_{nl}(-x, -y, -z) = R_{nl}(x, y, z)$$

结论： l 为奇数时是奇宇称， l 为偶数时是偶宇称。

球坐标表示的概率密度

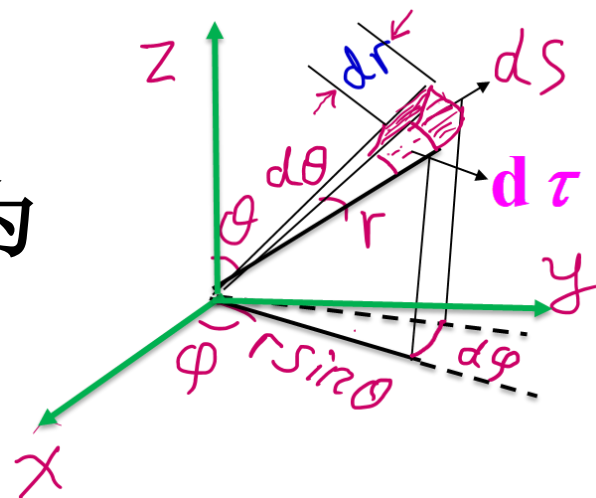
$|\psi|^2$ 代表几率密度

电子出现在 $d\tau$ 体积元中的几率为

$$|\psi|^2 d\tau = |\psi|^2 dr dS$$

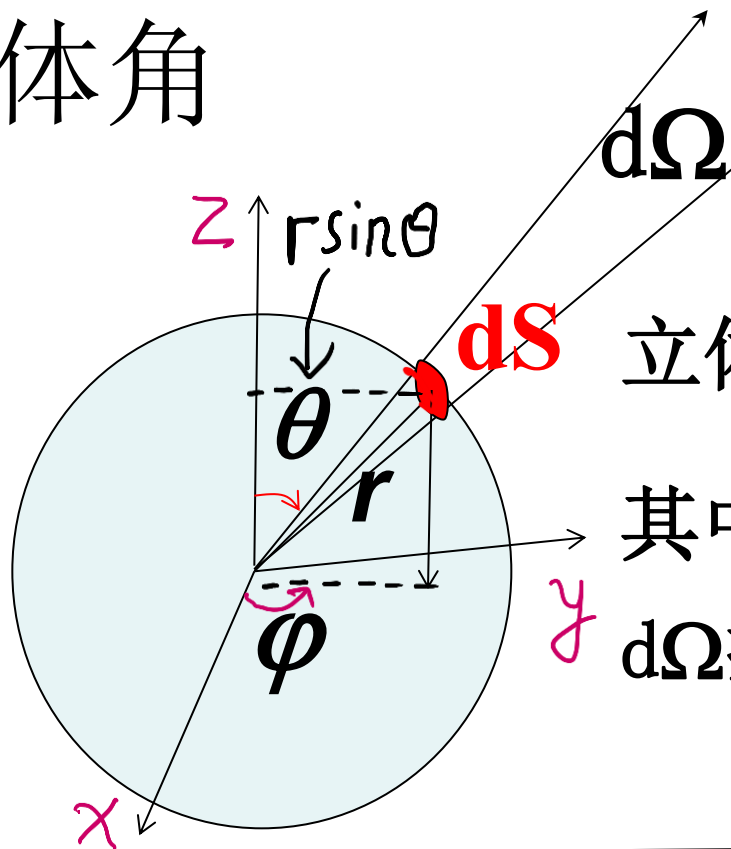
体元： $d\tau = dr dS$

面元： $dS = (r d\theta)(r \sin \theta d\varphi) = r^2 \sin \theta d\theta d\varphi$



下面需要分析角向和径向分布。

立体角

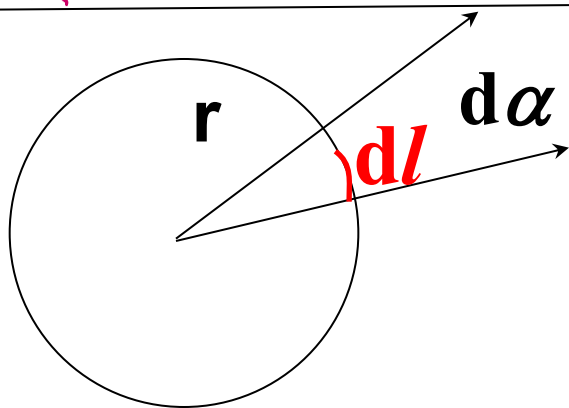
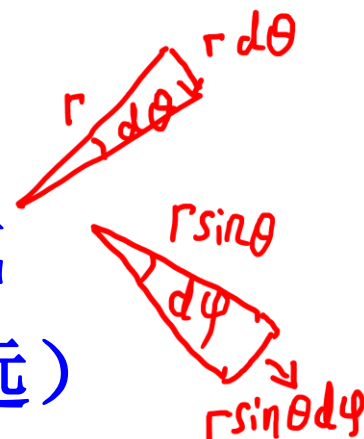


$$dS = (r d\theta)(r \sin \theta d\varphi)$$

$$\text{立体角 } d\Omega \equiv \frac{dS}{r^2} = \sin \theta d\theta d\varphi$$

其中 r 任选

$d\Omega$ 指图中锥形区域
(延伸至无穷远)



$$\text{类似于“平面角” } d\alpha = \frac{dl}{r}$$

两条射线所夹的区域

$$dS = r^2 d\Omega \quad d\tau = dS dr = r^2 d\Omega dr$$

电子出现在 $d\tau$ 体积元中的几率为

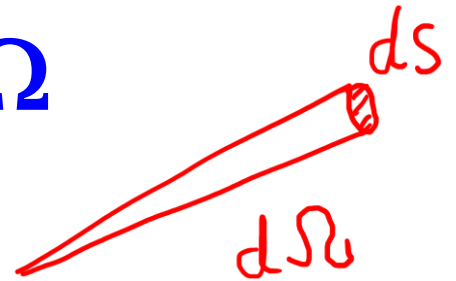
$$|\psi|^2 d\tau = R_{nl}^2(r) |Y_{lm}(\theta, \varphi)|^2 r^2 dr d\Omega$$

$$= R_{nl}^2(r) r^2 dr \times |Y_{lm}(\theta, \varphi)|^2 d\Omega$$

径向几率密度

角向几率密度

$$(d\Omega = \sin \theta d\theta d\varphi)$$

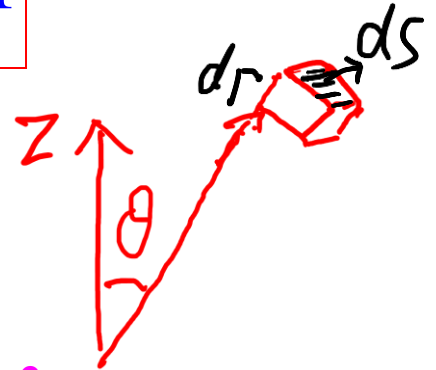


前面的径向和角向函数都已归一化，即

$$\int_0^\infty dr R_{nl}^2(r) r^2 = 1; \quad \int |Y_{lm}(\theta, \varphi)|^2 d\Omega = 1$$

于是在全空间积分

$$\begin{aligned} & \int_{\text{全}} |\psi(r, \theta, \varphi)|^2 d\tau \quad \text{体积元: } d\tau = \underline{ds dr} \\ & \quad \quad \quad = \underline{r^2 \sin \theta d\theta d\varphi dr} = r^2 dr d\Omega \\ & = \left[\int_0^\infty dr R_{nl}^2(r) r^2 \right] \times \left[\int |Y_{lm}(\theta, \varphi)|^2 d\Omega \right] \\ & = 1 \times 1 = 1 \quad (\text{全空间归一化}) \end{aligned}$$



练习：对以下氢原子基态波函数，
求径向几率密度

$$\psi_{100} = R_{10}(r) Y_{00}(\theta, \phi) = \sqrt{\frac{1}{\pi a^3}} \exp\left(-\frac{r}{a}\right)$$

$$\text{【其中 } R_{10}(r) = \sqrt{\frac{4}{a^3}} \exp\left(-\frac{r}{a}\right), \quad Y_{00} = \sqrt{\frac{1}{4\pi}} \text{】}$$

答案：解法1：由 $R_{10}(r)$ 直接写出

$$W(r) = r^2 R_{10}^2(r) = \frac{4r^2}{a^3} \exp\left(-\frac{2r}{a}\right)$$

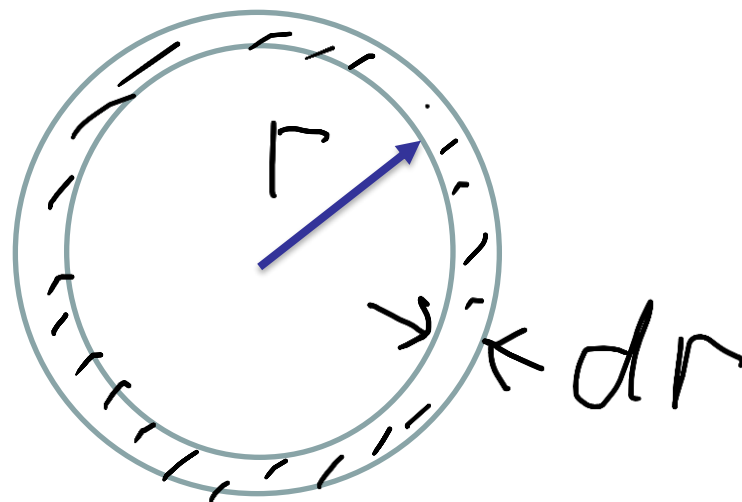
解法2：由 ψ_{100} ，对 Ω 积分

$$W(r) = \int_{\text{只对}\Omega\text{积分}} |\psi_{100}|^2 r^2 d\Omega = |\psi_{100}|^2 r^2 4\pi$$

$$= \frac{1}{\pi a^3} \exp\left(-\frac{2r}{a}\right) r^2 4\pi$$

$$= \frac{4r^2}{a^3} \exp\left(-\frac{2r}{a}\right)$$

$$\psi_{100} = \sqrt{\frac{1}{\pi a^3}} \exp\left(-\frac{r}{a}\right)$$

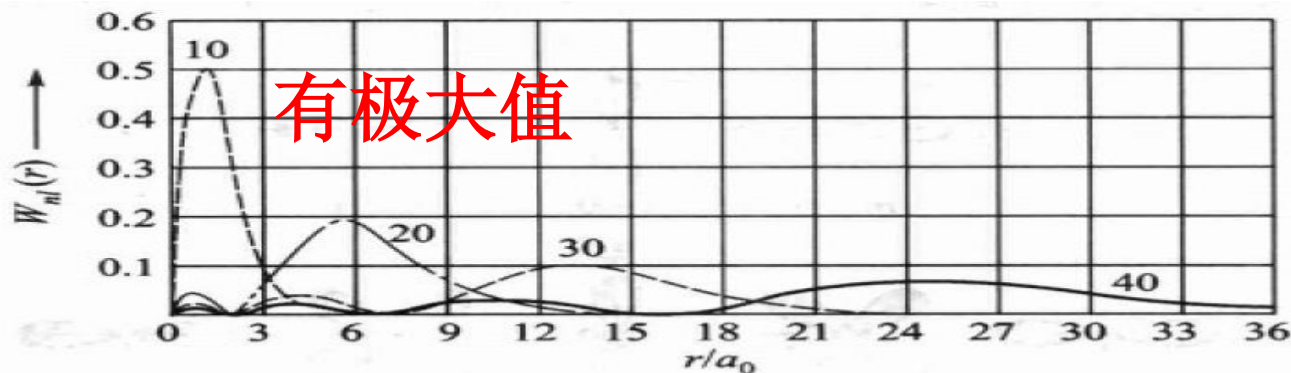


两种解法结果相同

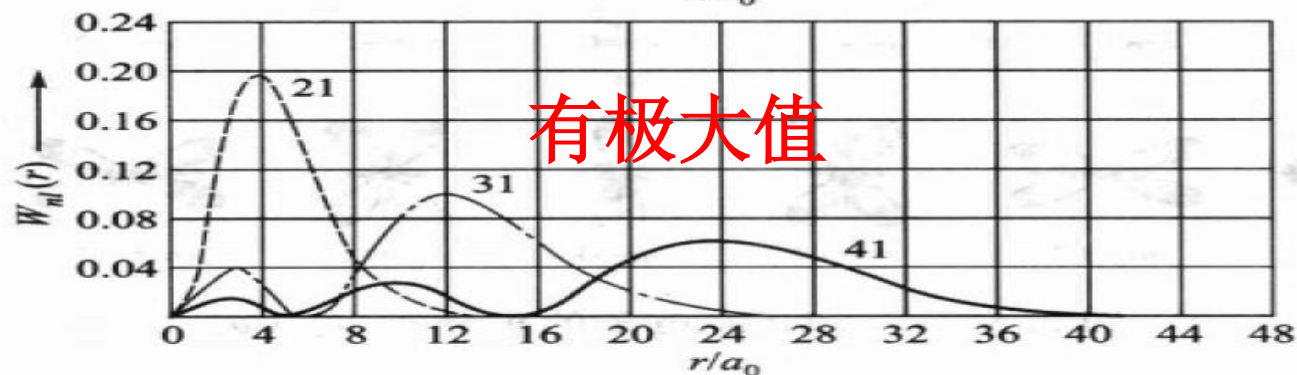
$$W(r) = R_{nl}(r)^2 r^2$$

径向几率密度随 r 的变化曲线

$W(r)$



$W(r)$



$W(r)$

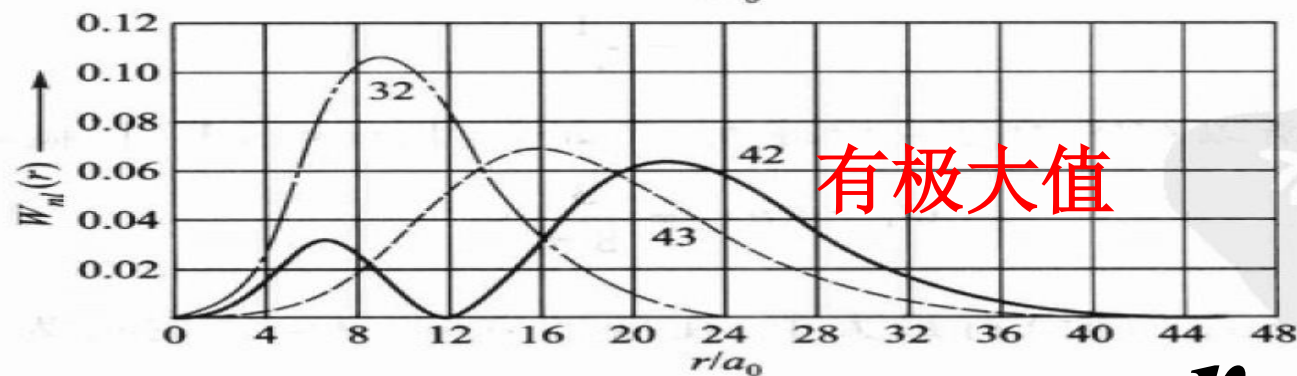


图 3.4 $W_{nl}(r)$ 与 r 的函数关系

r

角向几率 密度分布

球极坐标

注意：代表方向
分布（而不是轮廓）

$$|Y_{30}(\theta, \varphi)|^2$$

$$|Y_{31}(\theta, \varphi)|^2$$

$$|Y_{20}(\theta, \varphi)|^2$$

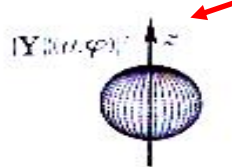
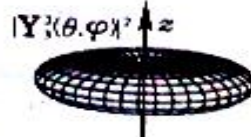
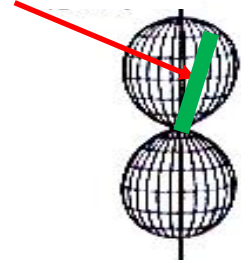
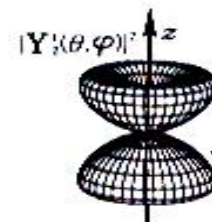
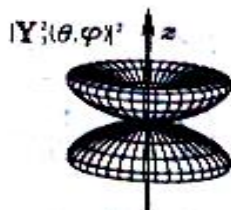
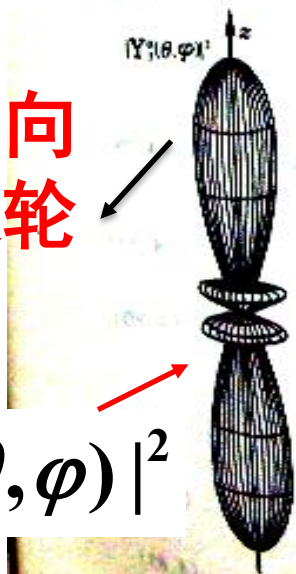
$$|Y_{10}(\theta, \varphi)|^2$$

$$|Y_{00}(\theta, \varphi)|^2$$

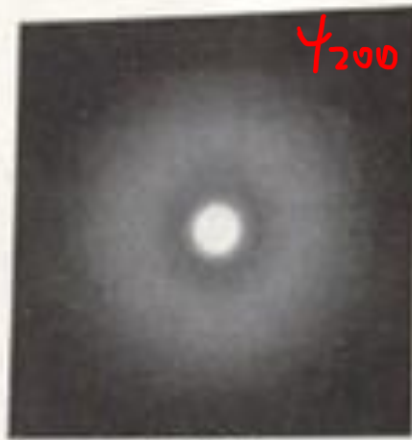
$$|Y_{11}(\theta, \varphi)|^2$$

$$|Y_{22}(\theta, \varphi)|^2$$

$$|Y_{21}(\theta, \varphi)|^2$$



$$|\psi_{nlm}|^2$$



(2,0,0)



(3,1,0)



(4,0,0)



(4,1,0)



(4,2,0)



(4,3,0)

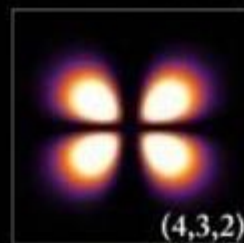
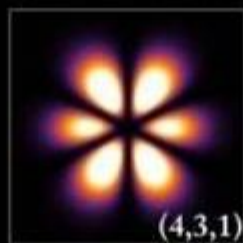
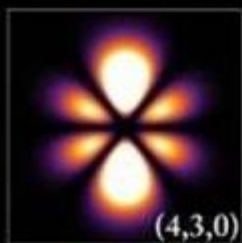
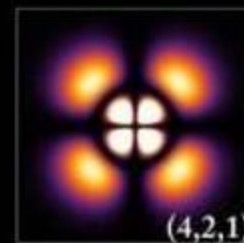
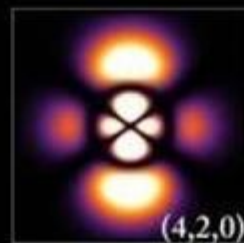
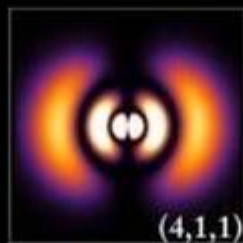
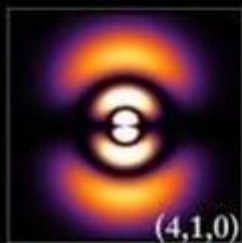
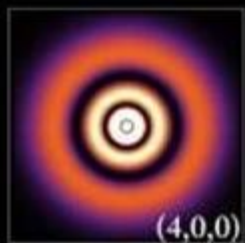
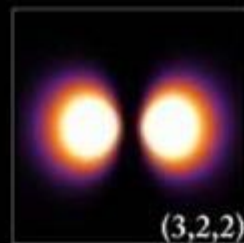
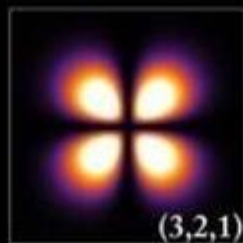
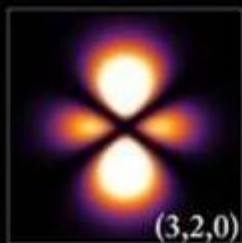
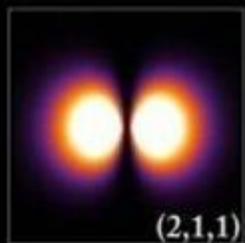
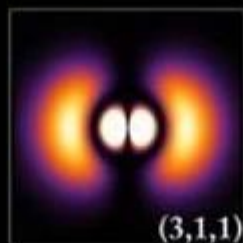
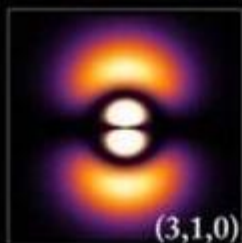
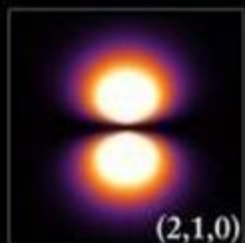
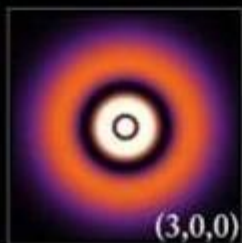
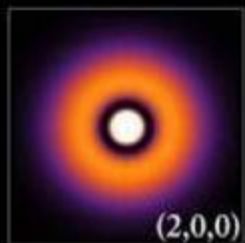
FIGURE 4.5: Density plots for the hydrogen wave functions (n, l, m) . Imagine each plot to be rotated about the (vertical) z axis. Printed by permission using "Atom in a Box," v1.0.8, by Darger Research. You can make your own plots by going to the Web site <http://darger.com>.

几率
密度
空间
分布

Hydrogen Wave Function

Probability density plots.

$$\psi_{nlm}(r, \vartheta, \varphi) = \sqrt{\left(\frac{2}{na_0}\right)^3 \frac{(n-l-1)!}{2n[(n+l)!]}} e^{-\rho/2} \rho^l L_{n-l-1}^{2l+1}(\rho) \cdot Y_{lm}(\vartheta, \varphi)$$



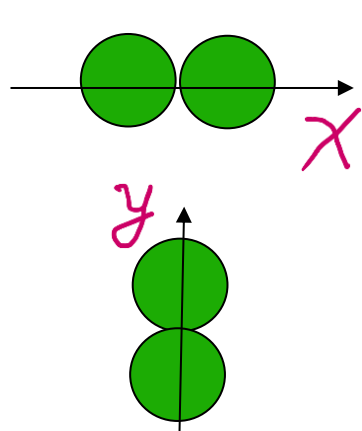
$$|\psi_{nlm}|^2$$

知乎 @Yang

*思考题:

角动量 L_x 和 L_y 的本征函数是否可以由 L_z 本征函数（球谐函数）线性组合而形成？ 如何组合？

提示： 例如可以考虑如下组合

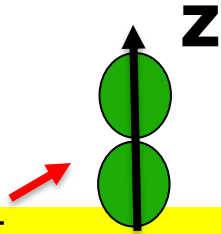


$$\frac{1}{\sqrt{2}}(Y_{11} - Y_{1-1})$$

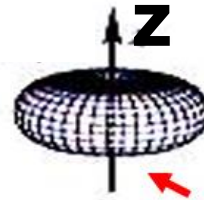
$$\frac{1}{\sqrt{2}}(Y_{11} + Y_{1-1})$$

可以证明它们分别是 L_x 和 L_y 的零本征值对应的本征函数。

附：思考题答案：



$$Y_z \equiv Y_{10} = \sqrt{\frac{3}{4\pi}} \cos \theta = \sqrt{\frac{3}{4\pi}} \frac{z}{r}$$



$$Y_{1,\pm 1} = \mp \sqrt{\frac{3}{8\pi}} \sin \theta (\cos \phi \pm i \sin \phi)$$

线性组合： $Y_x \equiv \frac{1}{\sqrt{2}} (Y_{1,1} - Y_{1,-1}) = -\sqrt{\frac{3}{4\pi}} \sin \theta \cos \phi = -\sqrt{\frac{3}{4\pi}} \frac{x}{r}$

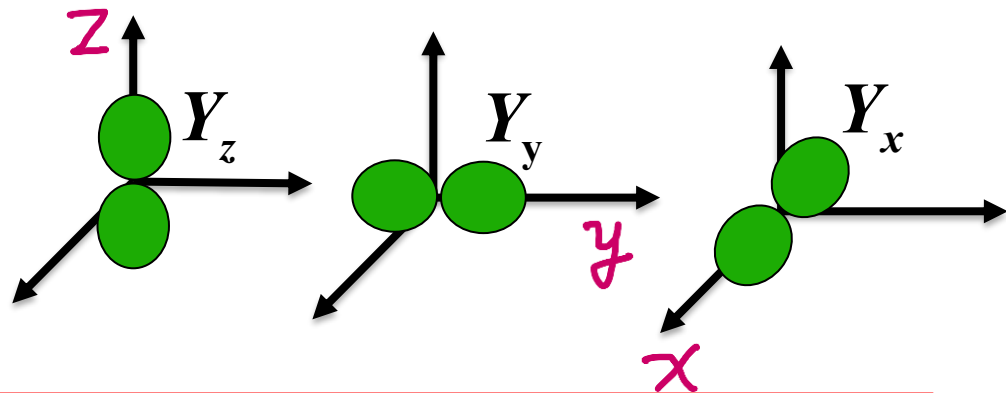
线性组合： $Y_y \equiv \frac{1}{\sqrt{2}} (Y_{1,1} + Y_{1,-1}) = -\sqrt{\frac{3}{4\pi}} \sin \theta \sin \phi = -\sqrt{\frac{3}{4\pi}} \frac{y}{r}$

$$\hat{L}_x = i\hbar \left(\sin \varphi \frac{\partial}{\partial \theta} + \cot \theta \cos \varphi \frac{\partial}{\partial \varphi} \right)$$

$$\hat{L}_y = -i\hbar \left(\cos \varphi \frac{\partial}{\partial \theta} - \cot \theta \sin \varphi \frac{\partial}{\partial \varphi} \right)$$

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \varphi}$$

附：思考题答案：



$$\hat{L}_z Y_z = \hat{L}_z Y_{10} = -i\hbar \frac{\partial}{\partial \varphi} \sqrt{\frac{3}{4\pi}} \cos \theta = 0$$

$$\hat{L}_x Y_x = -2i\hbar \sqrt{\frac{3}{8\pi}} (\sin \varphi \frac{\partial}{\partial \theta} + \cot \theta \cos \varphi \frac{\partial}{\partial \varphi}) \sin \theta \cos \phi = 0$$

$$\hat{L}_y Y_y = 2\sqrt{\frac{3}{8\pi}} i\hbar (\cos \varphi \frac{\partial}{\partial \theta} - \cot \theta \sin \varphi \frac{\partial}{\partial \varphi}) \sin \theta \sin \phi = 0$$

(得证)

小结

• 氢原子能量本征态 ψ_{nlm}

主量子数 $n = 1, 2, 3, \dots \rightarrow E_n = \frac{E_1}{n^2}$

角量子数 $l = 0, 1, \dots, n-1, \rightarrow L^2 = l(l+1)\hbar^2,$

磁量子数 $m = l, l-1, \dots, -l, \rightarrow L_z = m\hbar.$

简并度 $g_n = \sum_{l=0}^{n-1} (2l+1) = n^2.$

谱项(原子物理中的记号)

- 在原子物理中, 把能级 (态) 记为 nl , 称为**光谱项**, 其中 n 为主量子数, l 为角量子数, $l=0, 1, 2, 3\dots$ 分别用S, P, D, F...表示。

例如 2p 表示 $n=2, l=1$

不考虑精细结构时能量 E 只与主量子数 n 有关。

补充：电流密度

对定态 ψ_{nlm} , 可证电流密度 $\mathbf{J}_e = J_{e\varphi} \vec{e}_\varphi$

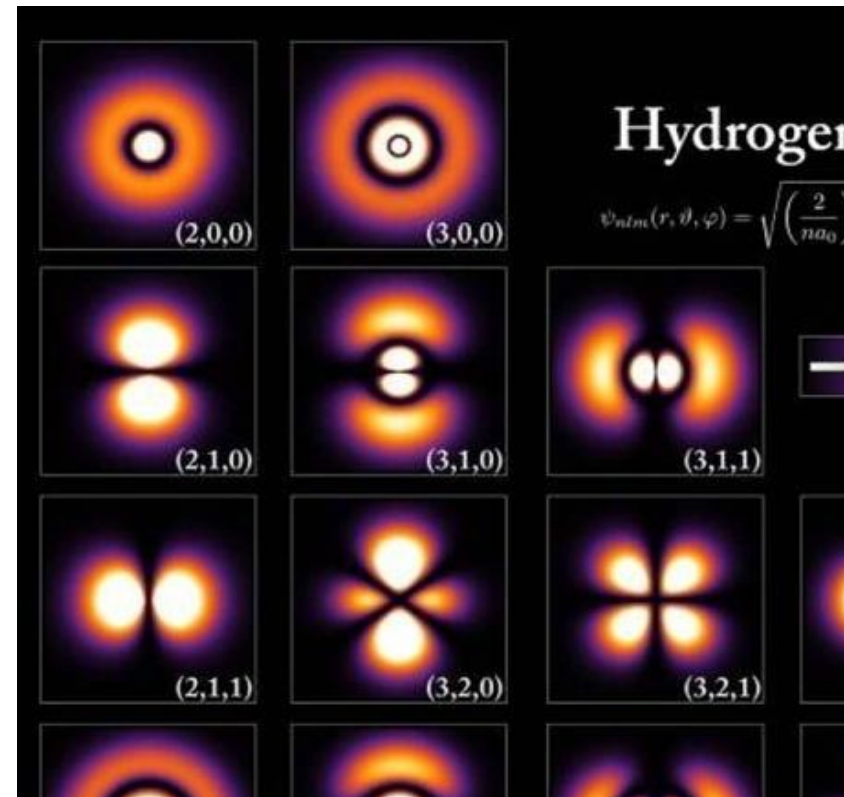
$m = 0$ 的态 $J_{e\varphi} = 0$,

$m \neq 0$ 的态 $J_{e\varphi} \neq 0$, 有磁矩, 体现了 **m 为磁量子数**

$$\mathbf{J}_{er} = \mathbf{J}_{e\theta} = 0$$

$$J_{e\varphi} = -\frac{e\hbar m}{\mu r \sin \theta} |\psi_{nlm}|^2$$

见作业



作业1:

- p. 91, #3. 3. 

3.3 证明氢原子中电子运动所产生的电流密度在球极坐标中的分量是

$$J_{er} = J_{e\theta} = 0$$

$$J_{e\varphi} = \frac{e\hbar m}{\mu r \sin \theta} |\psi_{nlm}|^2$$

• 提示 $\vec{J}_e = (-e) \frac{i\hbar}{2\mu} (\psi \nabla \psi^* - \psi^* \nabla \psi).$

$$\nabla = \vec{e}_r \left(\frac{\partial}{\partial r} \right) + \vec{e}_\theta \frac{1}{r} \left(\frac{\partial}{\partial \theta} \right) + \vec{e}_\varphi \frac{1}{r \sin \theta} \frac{\partial}{\partial \varphi}$$

$$\psi_{nlm}(r, \theta, \varphi) = R_{nl}(r) Y_{lm}(\theta, \varphi)$$

$$= R_{nl}(r) N_{lm} P_l^m(\cos \theta) \exp(im\varphi),$$

实数

$$\rightarrow J_{er} = J_{e\theta} = 0$$

$$J_{e\varphi} = - \frac{e\hbar m}{\mu r \sin \theta} |\psi_{nlm}|^2$$

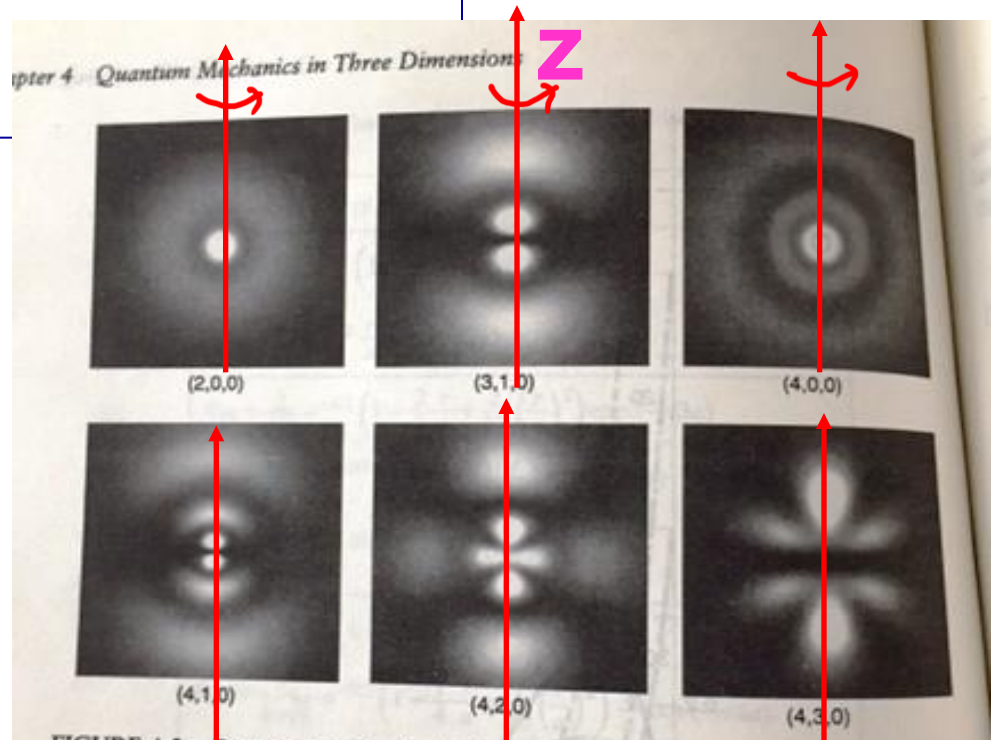
思考题

- 为什么波函数不是球对称的？
(自然界中的原子有方向性吗？)

$$\hat{L}_z \psi = -i\hbar \frac{\partial}{\partial \phi} \psi \Rightarrow \psi = R(r) P(\theta) e^{im\phi}$$

$|\psi_{nlm}|^2$ 与 ϕ 无关

分析：以上电子云
是 L_z 本征态。



*关于电子云的方向性

$$Y_{00} = \sqrt{\frac{1}{4\pi}} \text{(球对称)}$$
$$\left[\begin{array}{l} Y_{10} = \sqrt{\frac{3}{4\pi}} \cos \theta \\ Y_{1,1} = -\sqrt{\frac{3}{8\pi}} \sin \theta \exp(i\phi) \\ Y_{1,-1} = \sqrt{\frac{3}{8\pi}} \sin \theta \exp(-i\phi) \end{array} \right] \text{非球对称}$$

$$\sum_m |Y_{1m}(\theta, \phi)|^2 = \frac{3}{4\pi} \quad (\text{常数})$$

一般: $\sum_m |Y_{lm}(\theta, \phi)|^2 = \text{常数}$, 有何启示?

电子云空间分布图 (绕z轴旋转)

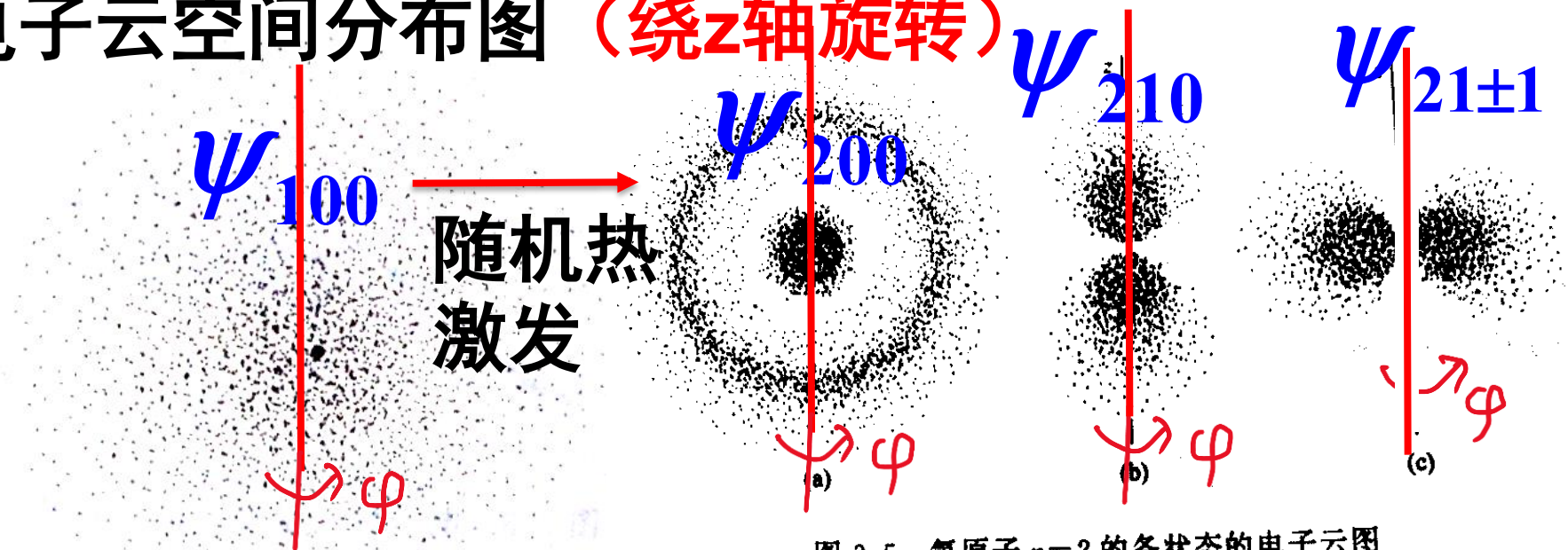


图 9-2 氢原子 $n=2$ 的各状态的电子云图

如果等概率 (随机) 热激发到 $n=2$ 的态, 则 3/03/16 17:58

$$\begin{aligned}
 & |\psi_{210}|^2 + |\psi_{211}|^2 + |\psi_{21-1}|^2 + |\psi_{200}|^2 \\
 &= |\mathbf{R}_{21}(\mathbf{r})|^2 \sum_m |Y_{1m}(\theta, \varphi)|^2 + |\mathbf{R}_{200}(\mathbf{r})|^2 |Y_{00}|^2 \\
 &= \mathbf{R}_{21}^2(\mathbf{r}) \frac{3}{4\pi} + \mathbf{R}_{200}^2(\mathbf{r}) \frac{1}{4\pi} \text{ (无方向性)}
 \end{aligned}$$

§ 3.4 本征函数系的一般性质

1. 正交性定理（重点）

- 正交性定理：同一个厄密（Hermitian）算符 $\hat{F}$ 的属于不同本征值的本征函数是彼此正交的。

正交的定义：
$$\int \psi_1^* \psi_2 \cdot d\tau = 0$$

证明

设 $\lambda_1 \neq \lambda_2$ $F\psi_1 = \lambda_1\psi_1, \quad F\psi_2 = \lambda_2\psi_2$

根据厄密性

$$\int \psi_1^* (\hat{F} \psi_2) \cdot d\tau = \int (\hat{F} \psi_1)^* \psi_2 \cdot d\tau$$

$$\Rightarrow \lambda_2 \int \psi_1^* \psi_2 \cdot d\tau = \lambda_1^* \int \psi_1^* \psi_2 \cdot d\tau$$

根据 $\lambda_1^* = \lambda_1$, 得 $(\lambda_2 - \lambda_1) \int \psi_1^* \psi_2 \cdot d\tau = 0$

因 $\lambda_1 \neq \lambda_2$, 所以 $\int \psi_1^* \psi_2 \cdot d\tau = 0$ (得证)

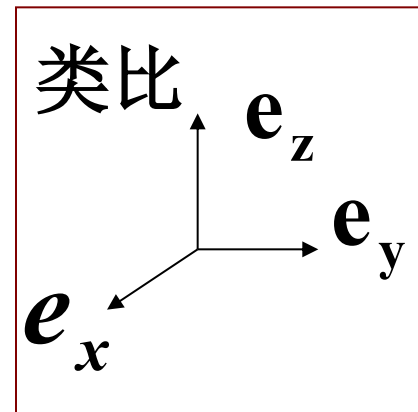
(不同本征值)

(正交)

本征函数正交“归一”性

本征值离散情况

“正交性”和“归一化”合写为：



$$\int \phi_k^*(\vec{r}) \cdot \phi_l(\vec{r}) d\tau = \delta_{kl}, \quad (k, l = 1, 2, \dots)$$

$$\delta_{kl} = \begin{cases} 0 & k \neq l \quad (\text{正交性}) \\ 1 & k = l \quad (\text{归一化}) \end{cases}$$

本征值连续情况 $\int \phi_{\lambda'}^*(\vec{r}) \cdot \phi_{\lambda}(\vec{r}) d\tau = \delta(\lambda - \lambda').$